Cyber-Resilient Recovery of Electric Vehicle Charging Under Uncertain Command Execution

Ali Sayghe and Ahmed Sultan
Department of Electrical and Electronic Engineering
Yanbu Industrial College, Royal Commission for Jubail and Yanbu
Yanbu, Saudi Arabia
sayghea@rcjy.edu.sa, Sultanba@rcjy.edu.sa

***Abstract*—Uncertain execution of electric vehicle charging commands can invalidate feeder capacity reallocation during recovery. We reserve capacity for charging caps consistent with causal execution evidence, including unresolved commands and expiry fallback. A trusted adapter supplies generation fencing, absolute expiry, and post-application certificates. A conditional proposition establishes cap containment and feasibility of selected affine network constraints. Nonlinear power-flow replay evaluates 3,980 episodes on IEEE 123-node and 13-node feeders with synthetic requests. Serialized and nonexpanding-supersession ledger variants exhibit no electrical violations in their evaluated cases. In the main impaired cases, supersession delivers 26.3% more energy than static quotas; a tuned feedback baseline violates limits in 22 of 240 cases. Supersession delivers 0.51% less energy than serialization, with a paired 95% interval of −3.23 to −1.69 kWh. A separately specified retry follow-up on new cohorts narrows this deficit to 0.17%. Field validation remains future work.**

***Index Terms*—Capacity reservation, charging recovery, cyber-physical security, distribution networks, electric vehicles.**

I. Introduction

Recovery of coordinated electric vehicle (EV) charging requires knowing which power limits can still execute. If a requested reduction is delayed, reallocating its capacity elsewhere can overload the feeder. A later measurement of low draw also need not exclude an unresolved increase or a higher fallback after profile expiry. These ambiguities concern executable limits as well as communication availability and charging demand.

For example, a 10-kW shared limit and active caps $(8, 2)$ cannot support reallocation to $(2, 8)$ until the first reduction is established. If a temporary 2-kW profile later expires to an 8-kW fallback, the same risk returns even after the reduction was observed. Recovery must account for both transitions.

Timed Command Interference links delayed commands and timeout fallback to EV scheduling violations; its proposed future defenses include execution verification and resilient coordination [1]. Related resilience mechanisms include capacity sharing during communication outages [2], feedback scheduling under uncertain charge acceptance [3], and distributed recovery from corrupted scheduling messages [4]. Networked predictive control already addresses uncertain applied inputs and timestamped actuator buffers [5]. These works establish the broader control and communication context. Here the specific object is the feeder capacity reserved for EV profiles that remain physically possible during recovery.

We construct a conservative ledger of active, unresolved, and expired profile histories under an explicit trusted adapter contract. Before transmission, each candidate command reserves its possible cap and subsequent fallback. Causal post-application evidence can release predecessors while retaining later unresolved generations. We compare two dispatch variants of this ledger: per-group serialization (B3) and nonexpanding supersession (P). Both share the same reservation accounting, scheduler, and information interface. The contribution is this charging-specific ledger, its conditional affine feasibility proposition, and a controlled evaluation of the variants' safety and service tradeoffs by nonlinear AC replay. The support-function calculation and generation fencing are established mechanisms. The adapter is a proposed interface, not a guarantee supplied by an ordinary Open Charge Point Protocol (OCPP) response [6].

II. System and Execution Model

Each index i denotes a feeder-connected charging phase group with actual nonnegative power p_i. A common scheduler greedily assigns group caps in order of earliest remaining vehicle departure, with group index breaking ties, subject to the positive-part affine constraint below. Within a group, local earliest-departure-first (EDF) allocation serves vehicles up to their constant acceptance limits, remaining requests, and the effective group cap. Requests, arrivals, departures, and acceptance limits are synthetic. An available booking service reveals a request at arrival; delivered energy becomes known only through delayed telemetry. This information interface is identical across policies and does not reveal future arrivals or physical completion privately.

A command carries group identity, monotonically increasing generation, constant cap q_i, and finite absolute validity interval $[s_i, e_i)$. The adapter has one authenticated authority, enforces the effective cap, and maintains a persistent known fallback f_i. Receiving a newer generation fences predecessor pending operations; acceptance alone does not establish application. Application follows the declared actuation lag and must finish before e_i. At absolute expiry, the temporary profile switches to fallback without extending its lease. The model uses exact clocks and zero tracking overshoot. Complete adapter state

survives modeled restart. This in-memory simulation establishes neither physical storage durability nor charger implementation validity.

Channel impairments include denial, delay, and delayed application retries after timeout or reconnection. They do not require reordered frames within an intact TCP stream. Neither power telemetry nor command acceptance eliminates an executable predecessor. A truthful application certificate identifies the group, generation, effective cap, application time, absolute end, and fallback. Its receipt is a separate delayed event.

Let $\mathcal{H}_i(t)$ conservatively retain cap histories consistent with initial state, issued commands, elapsed time, and certificates received by t. Let $c_i(h,s)$ denote the effective cap at time s in history h, including fallback, and let $r_i(I)$ be the power reserved for group i throughout a future interval $I \subseteq [t,\infty)$. The envelope and containment requirement are

$$U_i(I) = \sup_{h\in\mathcal{H}_i(t),\, s\in I} c_i(h,s), \qquad 0 \le p_i(s) \le U_i(I) \le r_i(I). \tag{1}$$

The inequalities apply to every $s \in I$. The ledger can overapproximate mutually incompatible histories. Its bound retains an older high cap through uncertain reduction, every unresolved possible increase, and fallback after each possible expiry, including fallback higher than the temporary cap.

Finite differences about a declared operating point give signed affine rows

$$Ap + Cd \le b, \qquad d \in [\underline{d}, \overline{d}], \tag{2}$$

where scalar d scales background demand. Rows represent upper and lower phase-voltage limits and source-rated transformer terminal-phase currents. The source files provide no line ampacities, so line thermal feasibility is outside this claim. Development margins tighten the affine limits; they are not certified bounds on linearization error. With $X^+ = \max(X,0)$ and $X^- = \min(X,0)$ entrywise, the robust reservation test is

$$A^+ r + C^+ \overline{d} + C^- \underline{d} \le b. \tag{3}$$

For each row, maximizing over the box $0 \le p \le r$ assigns $p_i = r_i$ to positive charging coefficients and zero to negative ones; maximizing the background term selects the corresponding interval endpoint. Thus (3) is the exact rowwise box counterpart of (2), although the box can be conservative relative to correlated physical histories.

III. Reservation Filter

At each control event, the filter processes delivered evidence and advances its ledger clock. It forms the nominal greedy targets from current bookings and received energy estimates, then considers reductions before increases without crediting the requested relief. For a candidate at group i, it constructs a prospective envelope containing the existing histories, the candidate cap during its lease, and fallback thereafter. All groups' current and candidate activation and expiry breakpoints partition the full finite commitment span. Feasibility is checked on every resulting interval, followed by the feasible persistent fallback. An optimization or simulation horizon ending cannot release capacity.

For the supersession variant P, an envelope-expanding command is admitted only when no unresolved operation expands the evidence-only envelope of that group. Otherwise, a new command is allowed only when its entire validity and fallback span is nonexpanding relative to the existing envelope. A smaller numerical cap alone is insufficient. This gate is recomputed after each certificate because an older reduction certificate may reveal that a later pending command is an expansion. The serialized variant B3 instead admits no new command while that group has an unresolved operation. Both variants permit different groups to progress concurrently.

The decision procedure is:

1) Incorporate causal certificates and advance known absolute expiries.
2) Compute nominal targets and order candidate reductions before increases.
3) Partition all affected future commitments, including fallback transitions.
4) Limit the candidate using (3) on every interval and apply the variant's unresolved-command gate.
5) Commit the prospective reservation before queueing the admitted command.
6) Query unresolved groups when available; release only histories excluded by truthful evidence or verified expiry semantics.

A certificate for generation g removes predecessor possibilities through g, but preserves all higher unresolved generations and their possible fallback transitions. Expiry can remove an executable temporary command without removing the fallback history it may have created. Reserve-before-send and retained state are modeled logically; deployment requires an atomic durable implementation.

Proposition 1: Assume initial containment, trusted cap enforcement and generation fencing, causal truthful evidence, bounded background demand, and coverage of all future commitments. If prospective reservations include each command and fallback, satisfy (3), and release histories only by the certificate and expiry rules above, actual charging satisfies the selected affine rows throughout every maintained feasible envelope.

Proof: Initially, the actual cap belongs to the retained history set. Reserving before send includes both nondelivery and possible application. Local application, fencing, and expiry follow covered histories; truthful evidence removes only excluded histories and retains later possibilities. Induction over these events therefore preserves $0 \le p \le r$. The rowwise maximization yielding (3) then bounds $Ap + Cd$ by b. ∎

This statement guarantees neither nonlinear AC feasibility nor minimum service under indefinite outages, and does not establish recursive feasibility of future service requests. AC outcomes are empirical. Proposition 1 applies to both ledger variants; neither is presumed to dominate in delivered energy. In the primary experiment, both query uncertain state without retransmitting lost commands.

IV. Evaluation Design

A. Electrical and Charging Models

The experiment uses the public IEEE 123-node and 13-node feeder definitions distributed with OpenDSS [7]. Six phase-connected EV groups are placed at buses 48.1, 76.2, 83.3, 67.1, 95.2, and 105.3 in the primary feeder. The 13-node check uses 632.1, 671.2, 675.3, 680.1, 645.2, and 611.3. These are unity-power-factor, medium-voltage equivalent loads; low-voltage cables, charger harmonics, battery aging, and protection dynamics are not modeled. Published background demand is first scaled to 80%, then multiplied by a scenario factor centered at 0.9 or 1.0 with a known ± 0.01 uncertainty interval. Regulators initialize at a 120-V target and their taps remain fixed during replay. Actual AC power flow is solved after each change in charging power or background demand.

Each 30-min episode contains 156 synthetic sessions with staggered arrivals and departures, heterogeneous requested energy, and constant 22-kW acceptance limits. Group caps are at most 300 kW. The 13-node check scales these powers and requests by $2/3$. Control and telemetry periods are 30 s; normal one-way delay is 2–2.9 s, and nominal actuation takes 1 s. Local EDF service integrates energy exactly between arrival, completion, departure, actuation, and expiry events. The persistent fallback is 65% of each group's initial feasible cap.

The affine model uses operating points of 100 kW/group on IEEE 123 and 50 kW/group on IEEE 13. Development tightening is respectively 0.005 and 0.002 p.u. for voltage, with a 0.98 loading limit on both feeders. Independent margin checks used 10 000 random power/background draws per feeder and loading center. The affine test admitted 9 110/6 860 draws on IEEE 123 and 9 629/5 999 on IEEE 13 at centers 0.9/1.0, with no admitted AC violation. This finite check informs the selected margins; it supplies no global AC guarantee. Replay limits remain 0.95–1.05 p.u. voltage and unity transformer terminal-phase current loading.

B. Channel Conditions and Comparators

The clean condition has only normal delay. T1 drops messages on three affected group links during a staggered 270–716 s window. T2 delays affected reductions until reconnection and adds 45 s to application-certificate delivery during the window. T3 delays affected increases until 180 s after reconnection, retaining possible late execution; certificates again incur the extra delay. T4 delays commands and queries around 900-s profile expiry, when a temporary low cap can revert to a higher fallback. The exact per-group offsets, queue rules, and absolute validity times are fixed in the companion configuration files. These are distinct stress conditions, not matched random-versus-selective attack distributions. The attacker controls the specified channel timing or availability, not authenticated command contents, the adapter, or its truthful post-application evidence. Selective cases grant observation of command direction at the messaging layer. This is an adversary-capability assumption; encrypted-transport compromise and charger exploitation are outside the model.

Seven policies share these physical and information interfaces. B0 reallocates as if requested caps immediately apply. B1 guards allocation using the latest measured and requested power with headroom, without retaining the full execution history. Its headroom was selected from 5%, 10%, and 20% on 50 development cases, choosing the smallest setting with zero observed electrical violations: 20%. B2 keeps the initial static feasible quotas. The serialized ledger variant B3 uses the execution ledger with per-group stop-and-wait and repeated state queries; an unresolved command blocks new commands until evidence or absolute expiry resolves it. Neither B3 nor P retransmits commands in the primary experiment. B4 uses a minimum-peak, deadline-energy linear program to form capacity-sharing requests, followed by the same serialization and feeder checks. It is inspired by dynamic capacity sharing [2], not a reproduction of that work's peer protocol or restoration objective. P permits the nonexpanding supersession of Section III. R gives P immediate truthful application certificates but retains forward-channel delay and common energy telemetry; it is not an omniscient service bound.

C. Pairing, Metrics, and Ablations

The primary study has 30 independently generated cohorts, five channel conditions, two loading centers, and seven policies: 2 100 episodes. The main impaired analysis contains 240 episodes per policy (Table I); the 60 clean episodes are separate. Another 840 episodes use ten new cohort seeds with 12 settings: certificate-delay extensions of 15/90/180 s, one/two/four affected groups, actuation lags of 0/5/15 s, and the three IEEE 13 T1–T3 checks. Finally, 80 ablation episodes remove pending-command accounting, remove fallback accounting, credit acceptance as application, or substitute a scalar aggregate budget for locational admission rows. Their matching full-P results come from five main cohorts at the higher loading center.

Service is energy delivered before departure; shortfall is requested energy minus that delivery. Safety metrics are the number of episodes with an actual AC limit violation, violation duration, and integrated excess voltage/loading. The primary paired P$-$B3 difference first averages the eight impaired conditions within each cohort. A 10 000-draw percentile bootstrap resamples the 30 cohort blocks, preserving all paired conditions, to obtain a 95% interval. Sensitivity and ablation estimates retain their ten- and five-cohort units, respectively. No timestep is an independent statistical sample. All controller settings and evaluation seeds were fixed after development and before evaluation outputs.

The primary ledger variants do not retransmit lost commands. A supplementary retry experiment, specified after the primary analysis and frozen before its outcomes were examined, uses 30 new cohorts, four impairments, two loads, and P/B3 with and without retries: 960 episodes. Both retry variants resend their latest unresolved generation every 60 s when no new command uses that group's remote procedure call (RPC) slot. Cap, generation, and absolute lease remain unchanged; duplicate

TABLE I
IMPAIRED MAIN RESULTS: 240 EPISODES PER POLICY. ENERGY AND SHORTFALL ARE MEANS; LOADING IS THE WORST TERMINAL-PHASE CURRENT RATIO.

Policy	Energy (kWh)	Shortfall (kWh)	Unsafe (/240)	Peak load (p.u.)
B0	546.2	222.9	192	1.120
B1	355.8	413.3	22	1.043
B2	380.3	388.8	0	0.980
B3	482.6	286.4	0	0.980
B4	466.9	302.2	0	0.977
P	480.1	288.9	0	0.983
R	481.9	287.2	0	0.983

receipt cannot reapply an accepted generation. Retries retain their original command direction for T2/T3 classification and replace a query on that tick. This follow-up is separate from the primary analysis and uses the same cohort-block inference.

V. RESULTS AND DISCUSSION

A. Electrical Outcomes and Service

All 3 980 episodes completed with converged AC solutions. In the main impaired cases, P delivers 480.1 kWh/episode, 99.84 kWh more than B2 (95% paired interval $[95.94, 103.87]$ kWh), a 26.3% increase. B2, B3, B4, P, and R have no observed electrical violations. P's worst voltage is 0.95487 p.u. and its largest terminal-phase current loading is 0.98299 p.u.; no cap-containment failure occurs. These are observed outcomes within the modeled conditions.

B0's greater delivery accompanies violations in 192 episodes, a minimum voltage of 0.93772 p.u., and a maximum loading of 1.11963 p.u. Even the clean condition produces 34/60 violating B0 episodes: ordinary latency already makes immediate capacity release unreliable. Additional impairment is therefore not the sole cause of this failure. B1's 20% margin, selected with zero development violations, yields 22/240 impaired and 2/60 clean violating cases. Sixteen impaired violations occur in T4, illustrating that development headroom does not certify expiry-aware capacity release.

B. Supersession Has a Service Cost

P does not dominate the query-only serialized variant. Its paired mean P−B3 difference is -2.48 kWh (95% interval $[-3.23, -1.69]$), or -0.51% relative to B3. Fig. 1 shows a negative effect in T1 and T2, an interval spanning zero in T3, and a small positive T4 effect of 0.135 kWh. In the clean condition, B3, P, and R have identical 489.74-kWh mean delivery. Thus the measured service penalty emerges when the policies interact with unresolved execution.

P issues more commands than B3 (86.3 versus 68.1 per impaired episode) but fewer queries (25.6 versus 49.1). Its integrated reserved-but-unused capacity is smaller, 16.89 versus 22.19 kWh-equivalent, without greater delivered energy. Reservation slack alone is therefore not a service objective. Fig. 2 presents the prespecified median-effect T1 example, selected across all 30 higher-loading cohorts. P delivers 9.43 kWh less than B3 in this case. R improves P's mean delivery to 481.9 kWh, but remains below B3; immediate application evidence is not a service upper bound for a greedy scheduler.

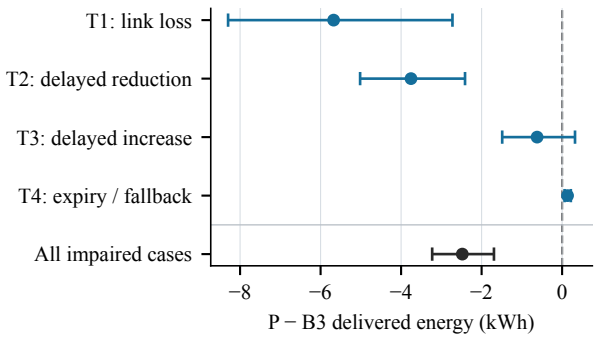


Fig. 1. Paired supersession-minus-serialization energy differences. The 95% intervals resample 30 cohort blocks; each condition retains both loading levels. Negative values favor B3.

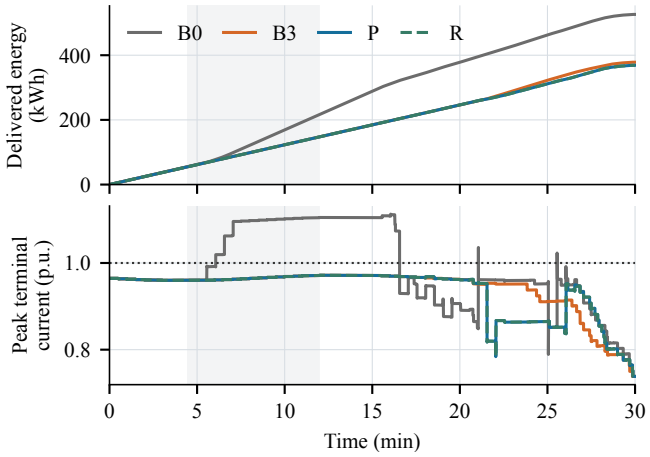


Fig. 2. Median-effect T1 example (cohort 1011, loading center 1.0). Curves show actual delivered energy and transformer terminal-phase loading. The shaded interval spans the staggered link outages. All trace replays reproduce their saved main-case metrics.

C. Supplementary Retry Comparison

Table II reports the separately frozen 960-episode follow-up. All four implementations have zero electrical violations or cap-containment failures. With retries enabled for both, $P_r - B3_r$ is -0.825 kWh (95% interval $[-1.137, -0.497]$), or -0.17%. On these same new cohorts, the query-only difference is -2.406 kWh. Thus equal retry capability narrows the aggregate gap without establishing a supersession service advantage. Retry-enabled P gains 0.400 kWh relative to retry-enabled B3 in T1, while losing 3.370 kWh in T2.

Retries are not guaranteed to improve this service metric: $B3_r - B3$ is -1.680 kWh, while $P_r - P$ is -0.099 kWh with an interval spanning zero. Retry attempts consume RPC opportunities otherwise available for state queries and alter subsequent greedy reallocations. The measured comparison supports the

TABLE II
SUPPLEMENTARY RETRY EXPERIMENT: 240 EPISODES PER POLICY. SUBSCRIPT R DENOTES 60-S SAME-GENERATION RETRIES. COMMANDS INCLUDE RETRANSMISSIONS.

Policy	Energy (kWh)	Unsafe (/240)	Commands (mean)	Queries (mean)
B3	484.7	0	67.9	49.0
P	482.3	0	85.1	26.1
$B3_r$	483.0	0	93.6	12.9
P_r	482.2	0	93.6	10.0

reservation accounting under both communication strategies, without recommending either as an optimal scheduling policy.

D. Sensitivity, Ablations, and Scope

P and B3 remain violation-free in all 120 sensitivity cases per policy, giving 420/420 when combined with their main cases. Longer evidence delay can change service: at a 180-s certificate-delay extension, P$-$B3 is -3.15 kWh with interval $[-5.56, -0.22]$. The 13-node checks also preserve electrical feasibility; they do not establish topology-independent guarantees. Across the 20 cases per ablation, omitting fallback accounting produces four violating episodes and replacing locational admission with an aggregate budget produces nine. Omitting pending-command accounting or crediting acceptance produces no AC violation in these sampled cases. These null outcomes do not establish that either omission is safe; they limit the empirical necessity claim for those mechanisms. Both safeguards remain assumptions of the formal result and are covered by execution-level regression tests.

On the recorded shared execution host, the median of episode-level median filter times is 0.773 ms for P and 0.619 ms for B3; observed maxima are 37.88 and 15.28 ms. These exclude AC replay and communication, and are not real-time implementation guarantees. Source, frozen scenarios, raw episode data, analysis scripts, and editable SVG figures are available at https://github.com/asayghe1/ev-charging-cyber-resilient-recovery. Synthetic demand, fixed acceptance limits, static feeder controls, and the trusted adapter restrict external validity. The implemented serialization variant provides greater aggregate service in the primary study; supersession is an option under the same reservations with condition-dependent service effects. This comparison does not establish an optimal production recovery policy.

VI. CONCLUSION

Execution-consistent reservations connect uncertain charging commands to feeder capacity that can actually be reused. Under the stated adapter contract, reserve-before-send, retained unresolved generations, and explicit expiry fallback preserve containment and the selected affine feasibility rows. The AC study finds no electrical violations for either reservation variant, while serialization delivers slightly more energy overall than nonexpanding supersession, including when both methods retry unresolved commands. This separates the safety contribution of execution accounting from the scheduling choice of when to replace a pending operation. Deployment requires validating the adapter contract, clock and tracking bounds, certificate integrity, and network-model accuracy on physical chargers. Field charging traces and low-voltage network detail remain necessary to assess operational generalization.

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